

Restaurant Orders System

A table-side ordering system: guests order from a device at their table in German, English or Arabic; the kitchen and the dining hall see every order live.

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1 The idea

In a restaurant today, an order passes through several people before it reaches the kitchen: the guest waits for a waiter, the waiter writes the order down and carries it back, then returns again to say it is ready. Every hand-off costs time and can lose detail.

This project replaces that chain with four connected screens. The guest orders directly from a device on the table and receives an order number. The kitchen sees the order the moment it is confirmed. A display in the dining hall shows which numbers are being prepared and which are ready. A management backend behind all of it defines the menu and controls every order's status.

2 The four screens

SCREEN	WHERE IT RUNS	WHAT IT DOES
Guest ordering	Tablet on the table, or the guest's own phone	Language choice, menu with photos and prices, meal details, cart with running total, payment method, confirmation with order number and live status.
Kitchen display	Wall monitor in the kitchen	Three columns — new, in preparation, ready. Each card shows the order number, table, items and notes, with a timer that turns amber then red. Two buttons: start, ready.
Hall status board	Screen in the dining area	Order numbers only, in two large columns: in preparation, and ready. No names, no prices.
Management backend	Office computer	Menu, raw materials, prices, tables, staff users, settings, and full control over the status of every order.

3 Features

Ordering

- Browse the menu by category, with photos and prices
- Meal detail: description, preparation method, ingredients
- Several sizes per meal, each with its own price
- Quantity control and a free-text note for the kitchen
- Running total, then subtotal, tax and total in the cart
- Choice of payment method before confirmation

Management

- Add, edit and delete raw materials, categories and meals
- Each meal: title, description, preparation method, ingredients, sizes, prices
- Recipes link each meal size to its raw materials
- Mark a meal unavailable — it disappears from every table at once
- Register tables and pair each one to its device
- Staff accounts with roles: admin, kitchen, waiter, cashier

- An order number, shown large, with a live status timeline

Languages

- German, English and Arabic
- Both interface labels *and* menu content are translated
- Arabic is fully right-to-left, not mirrored English
- Numbers and currency formatted per language

Configuration

- Currency code, symbol and symbol position
- Tax rate
- Default language
- Which payment methods are offered: cash, card, PayPal, cash desk

4 The order process

Every order follows one defined path. Each change of stage is recorded with a timestamp and the member of staff who made it, so the full history of an order can be reconstructed.



STAGE	MEANING	TRIGGERED BY
Draft	The guest is still adding items; not visible to the kitchen	Guest, on the table device
Submitted	Confirmed, payment method chosen, order number issued	Guest confirms the order
In preparation	The kitchen has started cooking	Kitchen screen
Ready	Finished; the number appears on the hall board	Kitchen screen
Served	Delivered to the table; leaves the active boards	Waiter or admin
Cancelled	Stopped before serving; flagged in red on the kitchen screen	Admin

5 The three user journeys

Guest

Welcome screen → choose language → browse menu → open meal detail, pick size and quantity, add a note → add to order → review the cart and the total → choose a payment method → confirm → receive the order number and watch its status until it is ready.

Kitchen staff

A new order appears in the *new* column with its number, table and items → press **Start** when work begins, and the order moves to *in preparation* → press **Ready** when the food is finished, and the number appears on the hall board → a cancelled order is flagged in red until acknowledged. A meal that has run out can be marked unavailable and vanishes from every table device.

Administrator

Define raw materials → create categories → create meals with their translations, sizes, prices and recipes → register tables and staff accounts → set currency, tax and payment methods → monitor all orders, filter by date and status, and move any order to the correct status when something goes wrong on the floor.

6 What makes it a substantial project

Beyond the screens, the parts worth defending in a review are these.

- **A state machine, not a status field.** Every status change goes through one place that validates the transition, sets the timestamps and writes an audit record. No part of the system may set a status directly.
- **Order numbers must be unique under load.** Two tables confirming at the same instant must not receive the same number, which makes number generation a concurrency problem, solved with a locked transaction rather than a counter.
- **Prices are calculated on the server, never on the device.** The total shown to the guest is recalculated and verified before the order is accepted.
- **Orders store a snapshot of what was sold.** Meal title, size and price are copied onto the order line, so changing a menu price tomorrow cannot rewrite yesterday's receipts.
- **Internationalisation and right-to-left layout** designed in from the start, with translated content stored per language so a missing translation can be found and filled.
- **Role-based access** so each of the four screens exposes only what its user is allowed to do.

7 Scope

BUILT IN THIS PROJECT	DELIBERATELY LEFT FOR A LATER VERSION
All four screens, complete	Live card and PayPal charging (the flow is built; real charging needs a merchant account)
Three languages with right-to-left support	Automatic stock deduction from recipes
Full menu and raw-material management	Kitchen ticket printing
Order lifecycle with audit history	Reporting and analytics
Configurable currency, tax and payment methods	Instant push updates in place of periodic refresh
Role-based staff accounts	
Installable on phones and tablets as an app	

Demonstration. The system can be shown running end to end: an order placed on a phone appears on the kitchen screen within seconds, and its number moves across the hall board as the kitchen advances it — all three screens visible side by side.